This procedure is applicable to below models:

V810 STD	√	S1	√	S2
V810 XXL		S1		S2
V810 S2EX				
V810 Mini				

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	06 Apr 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG
02	23 May 2017	Revise Step and Add Replace Procedure	CHIN CHEE YAN

Content

Removal Procedure

1. **Shutdown and completely power off machine**. Turn off the main air pressure regulator
2. **Power off the SSCA main power** # Importance.
3. Open front and rear access door
4. Disconnect 4 x connectors at Right Outer Barrier

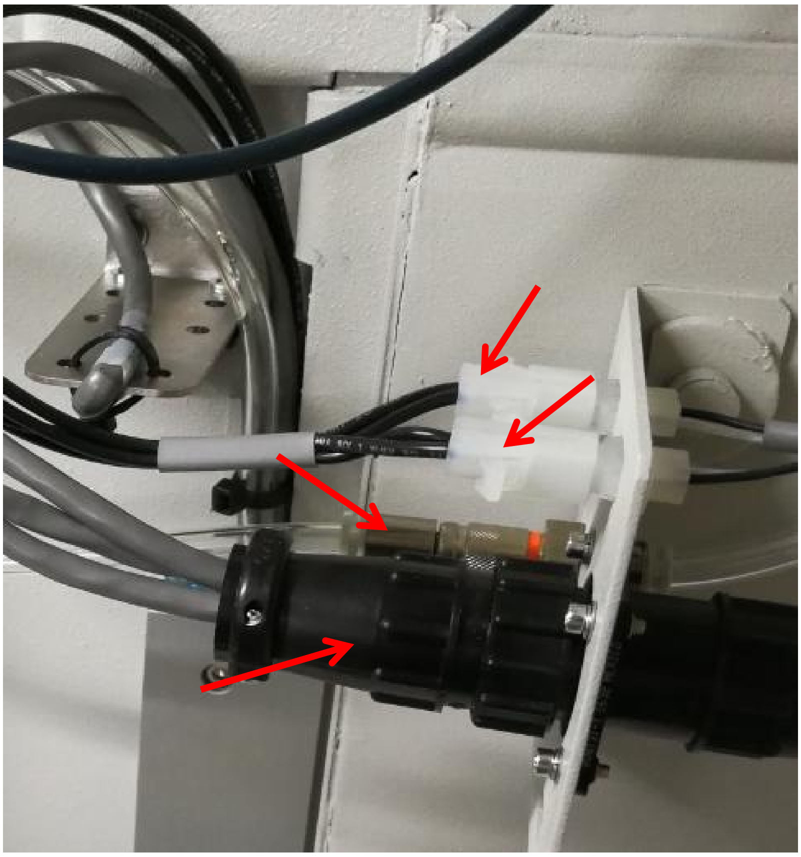


Figure 1 Outer Barrier Connector Removal

5. Remove 4 x Tamper Proof screws as below Figure 2

Figure 2 Right Outer Barrier Removal

6. Manually move the XY stag towards right outer barrier
7. Disconnect the Cable, Motor Stepper AWA Axis Adjustable Rail (P/N, 4XASC-A079) from Right AWA motor. (Remarks : Right Motor location define by Front Door Direction)

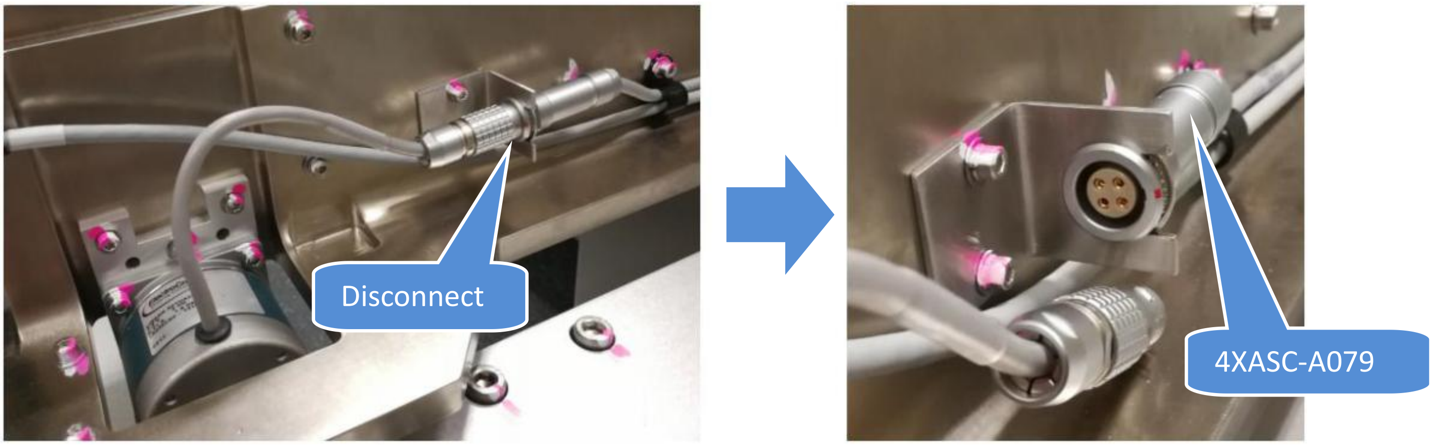


Figure 3 Right Stepper Motor Connector Removal

8. May refer to below diagram for Right Stepper Motor Cable Routing

Route : Right Stepper Motor -> AWA E-Chain -> X axis E-Chain -> Y axis Wide E-Chain -> SSCA

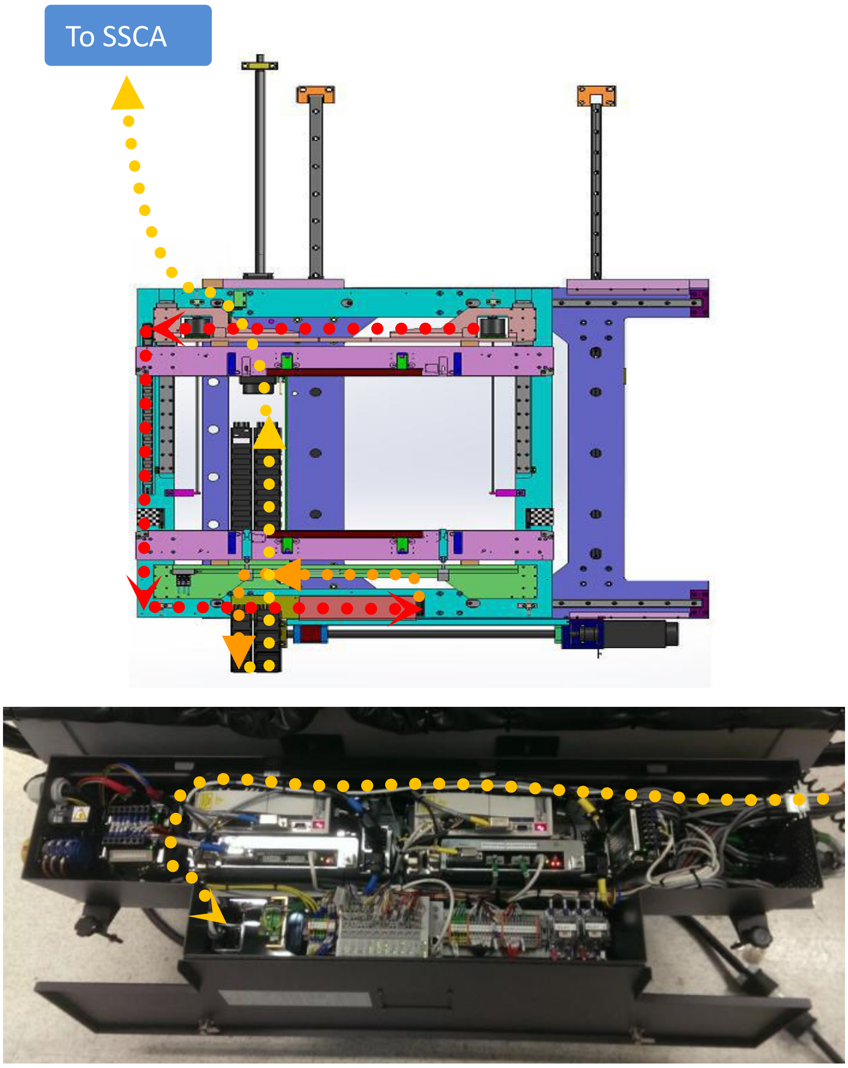


Figure 4 Right Stepper Motor Cable Routing

9. Loosen the lock nut at Cable, Motor Stepper, AWA Axis (4XASC-A079) from BRACKET, SINGLE PKG (P/N, 4XAXY-061) by using 17mm wrench.

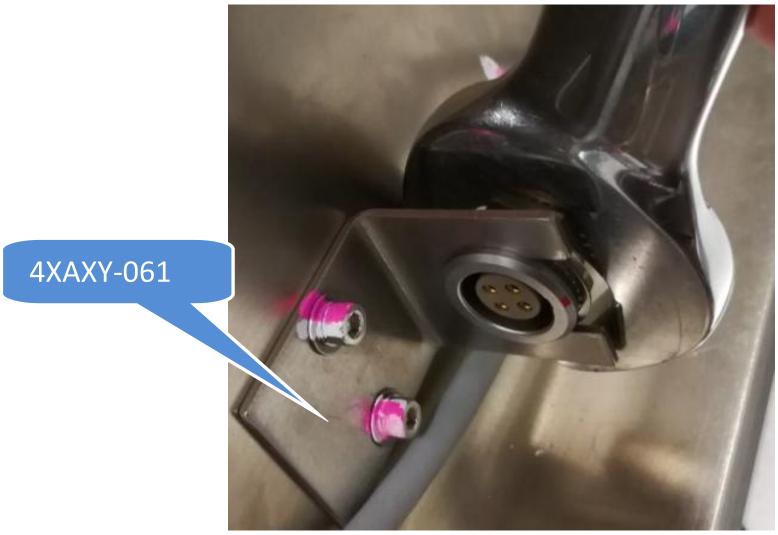


Figure 5 Lock nut Loosen

10. At the adjustable rail, loosen Cable Clamp and remove 4XASC-A079 from Cable Clamp. Remove cable tie as well

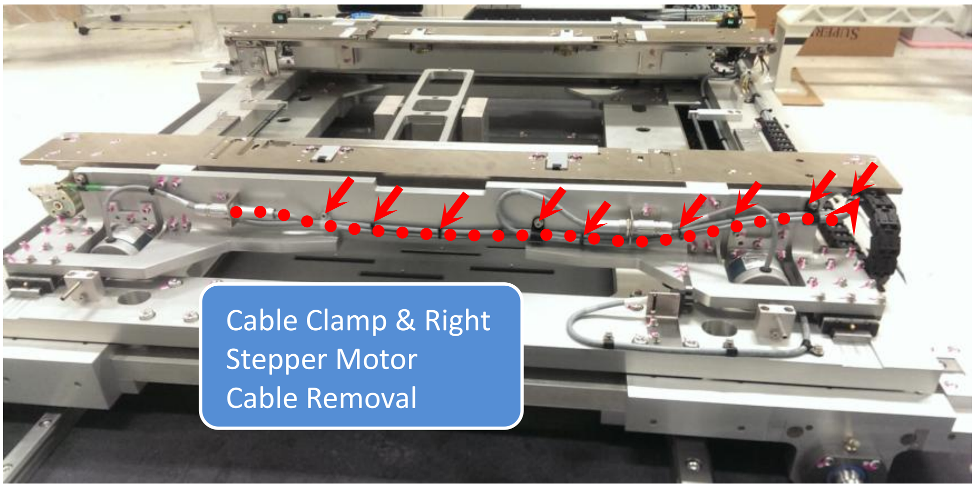


Figure 6 Cable Clamp & Right Stepper Motor Cable Removal

11. Loosen 4 x Screws and remove CHAIN BRACKET (P/N, 4XAXY-117) as below Figure 7

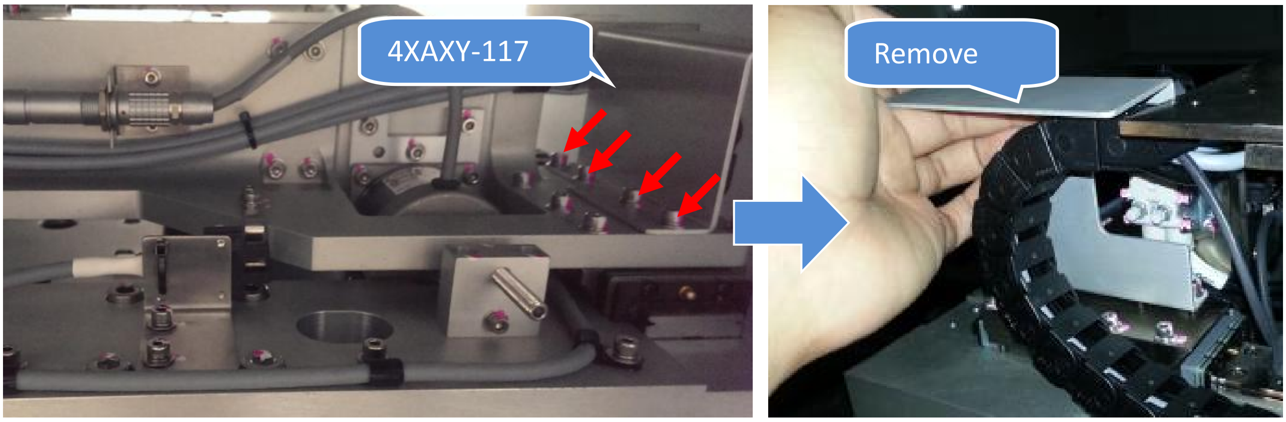


Figure 7 Chain Bracket Removal

12. Use flat head screwdriver to open AWA E-Chain Top Joint

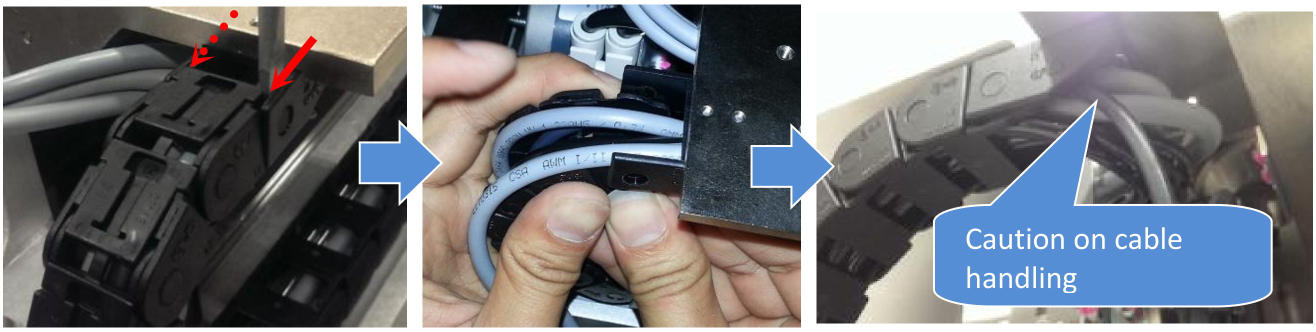


Figure 8 AWA Top Joint Removal

13. Use flat head screwdriver to open Slot Cover along the AWA E-Chain

Figure 9 E-Chain Slot Cover Open

14. Use flat head screwdriver to remove AWA E-Chain bottom joint as below Figure 10

Figure 10 E-Chain Bottom Joint Removal

15. Remove 4XASC-A079 from AWA E-Chain as below Figure 11

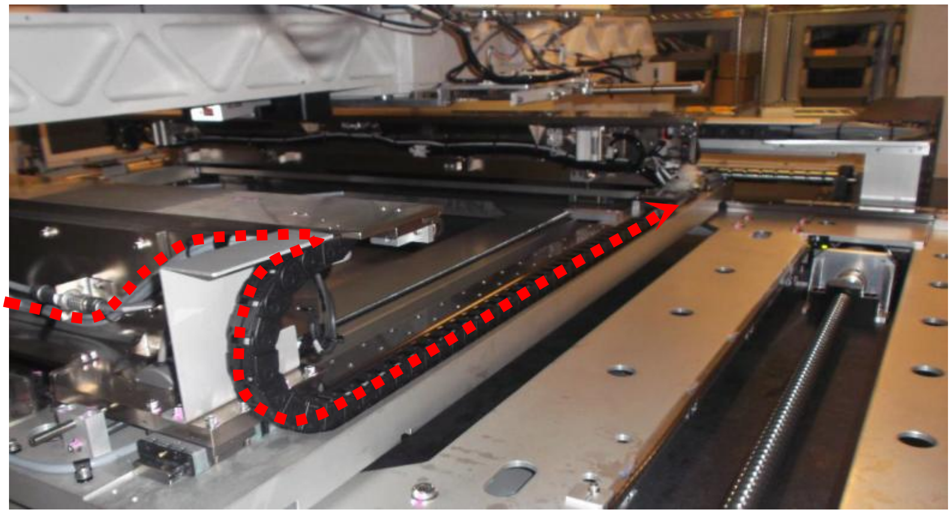


Figure 11 Stepper Motor Cable Removal (AWA E-Chain)

16. Loosen 2 x screws remove the SENSOR, REFLECTOR, E39-R (P/N, 2284-0052) as shown

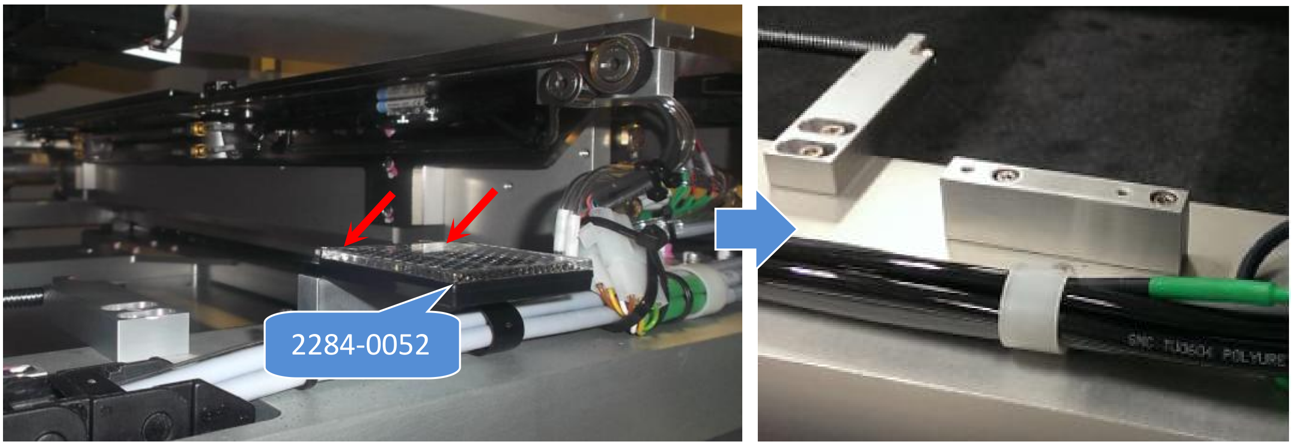


Figure 12 PCL Sensor Reflector Removal

17. Track remaining route and remove stepper motor cable. Remove necessary cable clamp & cable tie

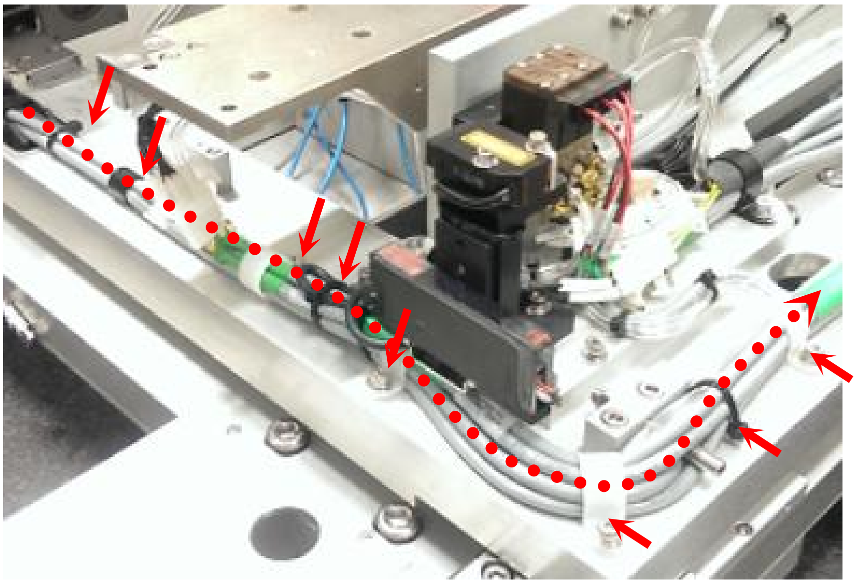


Figure 13

18. Gently lift up Motor Cables and loosen 2 x screws as shown.Open & remove the GUARD, E-CHAIN (P/N, 4XAXY-075).

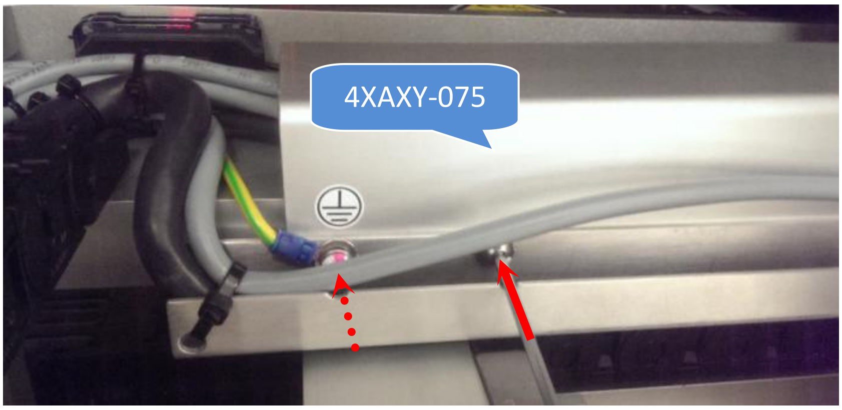


Figure 14 X Axis E-Chain Guard Plate

19. Use flat head screwdriver to open V810 S2 X-AXIS E-CHAIN (P/N, 4XAXYS-A174) end joint (top and bottom) as below Figure 15

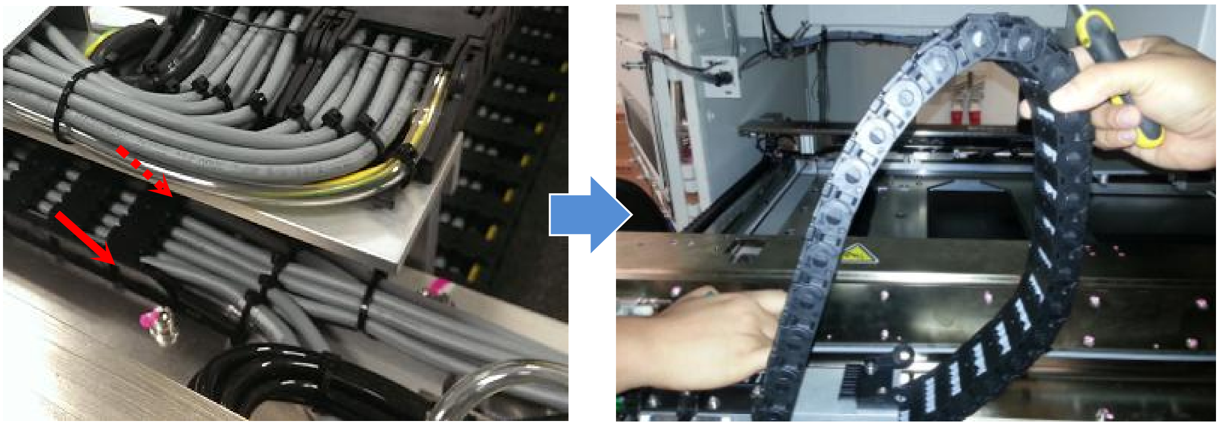


Figure 15 X Axis End Joint Removal

20. Use flat head screwdriver & open E-chain cover top cover along X axis E-Chain

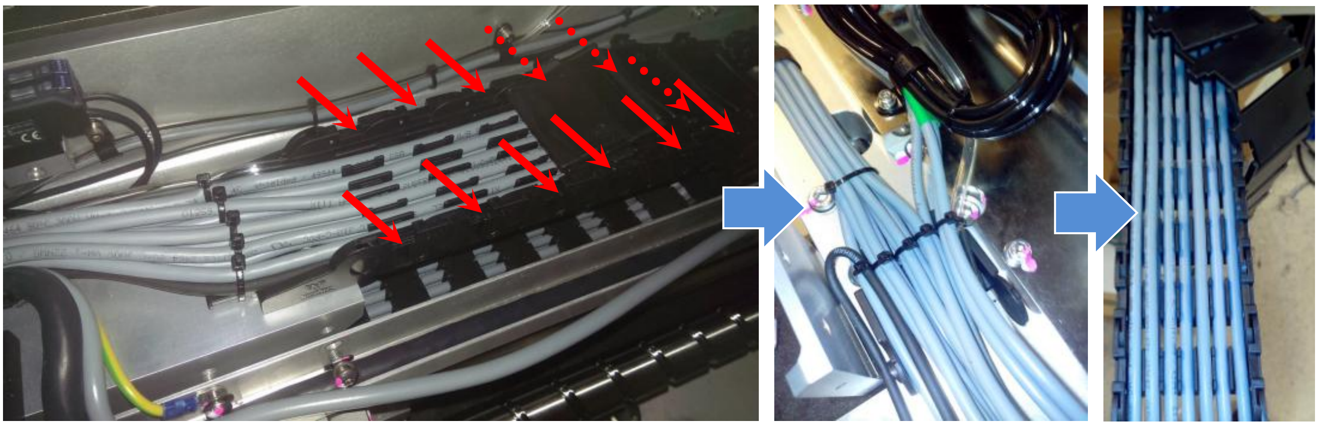


Figure 16 X Axis E-Chain Cover Removal

21. Remove 4XASC-A079 from X axis E-Chain as below Figure 17. Remove necessary cable tie

Figure 17 Stepper Motor Cable Removal (X Axis E-Chain)-Reference Picture Only

22. Track 4XASC-A079 cable route along Y axis Wide E-Chain.Use flat head screw driver to remove E-Chain Top cover along Y axis Wide E-Chain. Next,open Y axis Wide E-Chain bottom joint

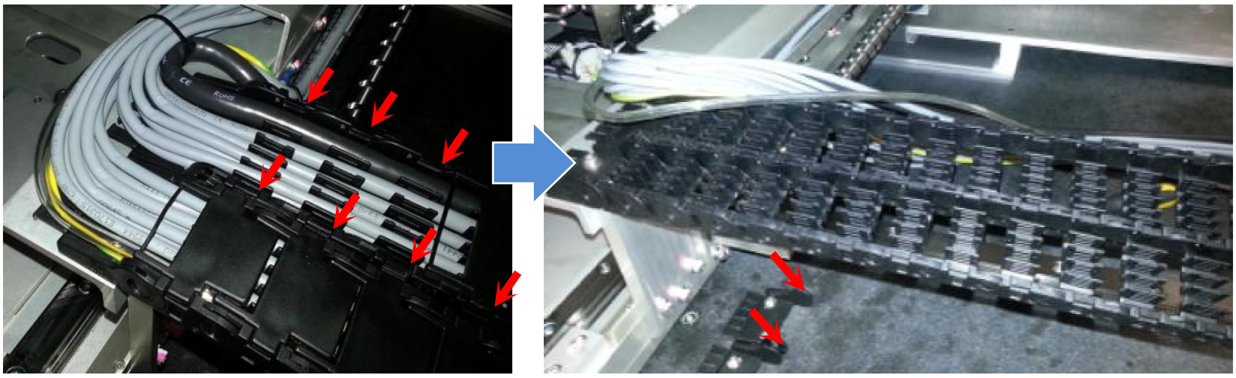


Figure 18 Stepper Motor Cable (Y Axis Wide E-Chain) Reference Picture

23. Remove 4XASC-A079 from Y axis Wide E-Chain. Remove necessary cable tie as well

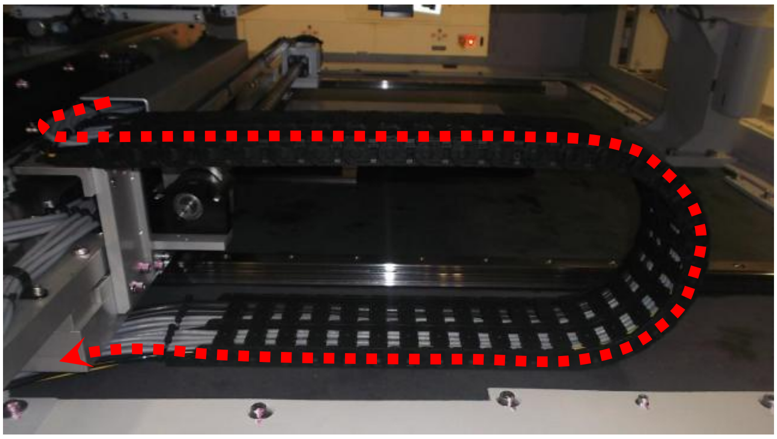


Figure 19 Stepper Motor Cable Removal (Y axis Wide E-Chain)

24. Track remaining route and remove 4XASC-A079 accordingly. Remove necessary cable tie.

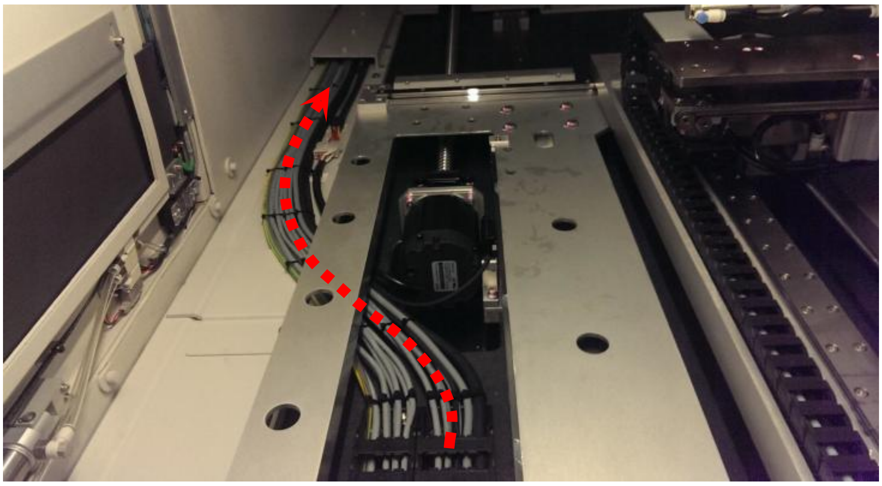


Figure 20 Stepper Motor Cable Removal

25. Open Rear side door & loosen 5 x Screws.Remove" Vertical Lead Shielding Assembly" and "Cable Exit Shielding Assembly" as below Figure 21

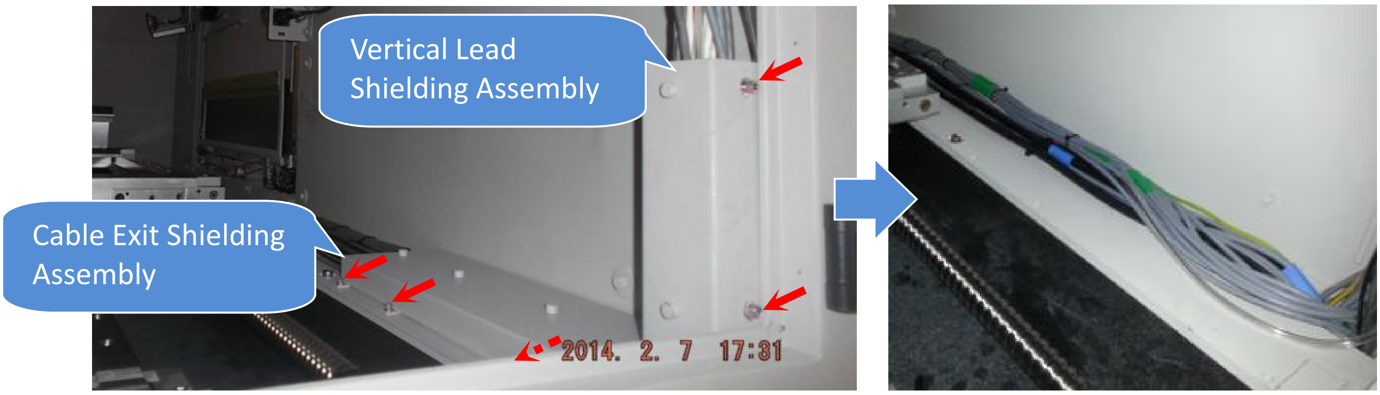


Figure 21 Shielding Assembly Removal

26. Gently remove 4XASC-A079 through the "Opening"

Figure 22

27. Gently remove 4XASC-A079 as shown. Remove necessary cable tie

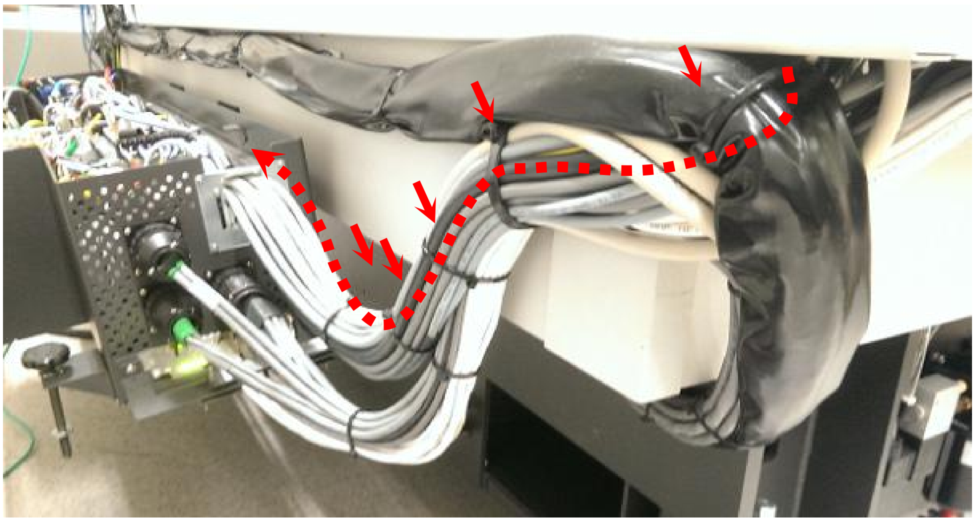


Figure 23 Stepper Motor Cable Removal

28. Loosen 2 x Screws.Remove MCA Lambda HWS 100 Body Holding Plate Assembly (P/N, 4XAMC-A045) as shown.

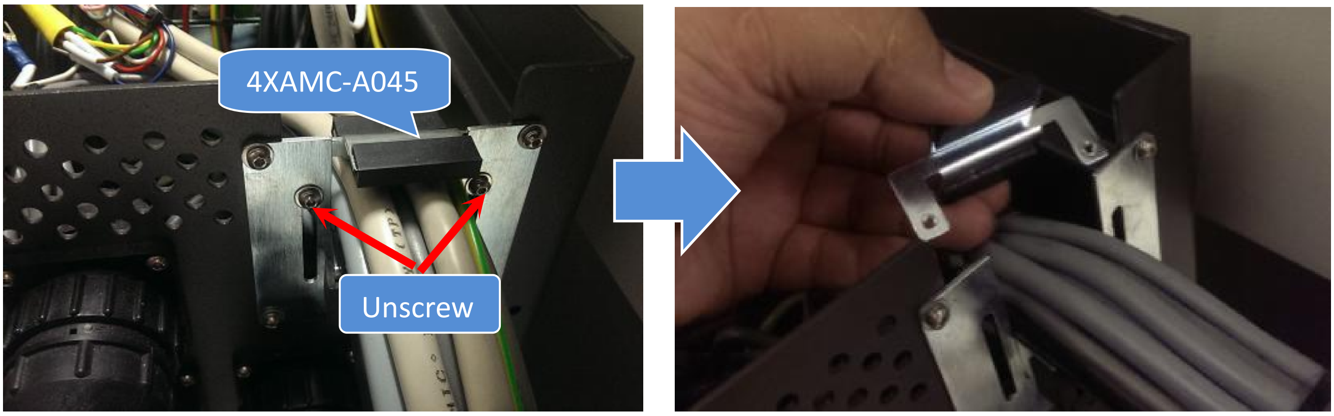


Figure 24 Holding Plate Removal

29. Track 4XASC-A079 route at SSCA.

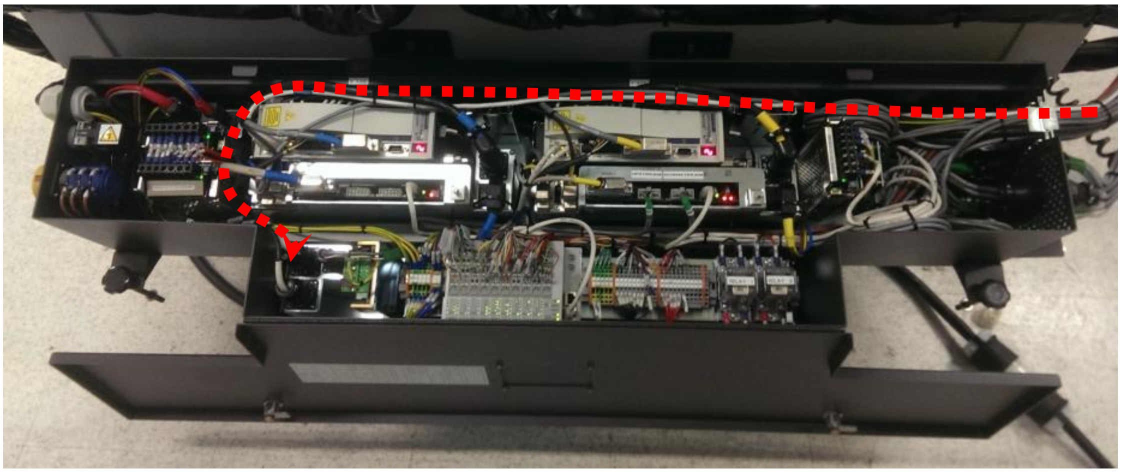


Figure 25 Stepper Motor Cable Removal (SSCA)

30. Track stepper motor cable end. Disconnect bulb head connector by refer to Cable Label and remove from stepper driver

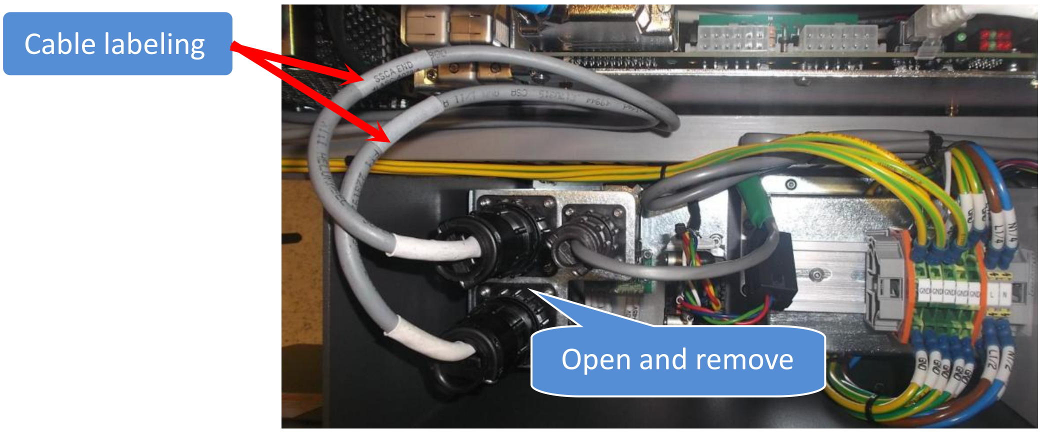
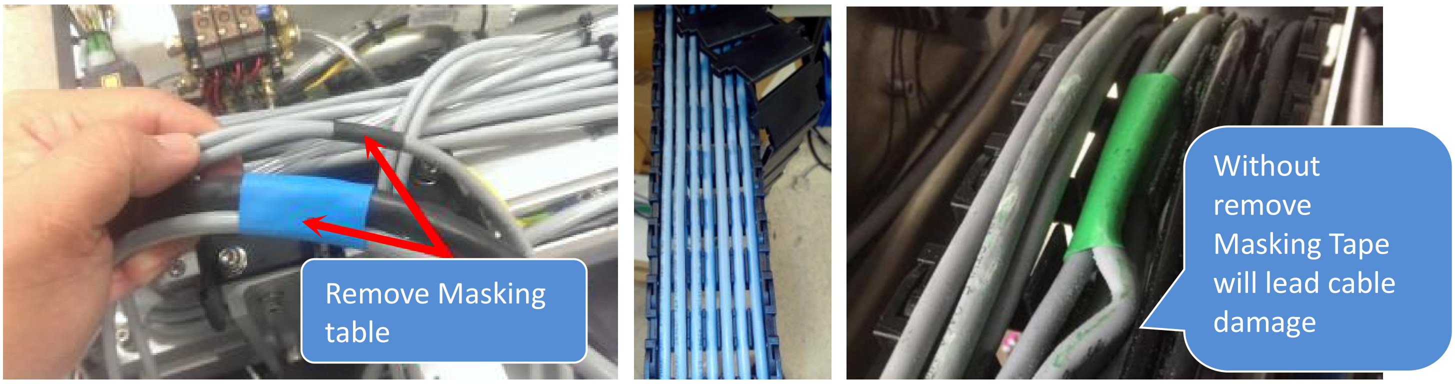


Figure 26 Stepper Motor Cable Removal (Stepper Driver)

Replacement Procedure

Caution : Cable within E-Chain where apply with Masking tape MUST REMOVE



Spare Cable Masking tape | Good routing | Bad routing

Figure 27

31. Prepare new CABLE, MOTOR STEPPER AWA AXIS (P/N, 4XASC-A079) as shown.

Figure 28

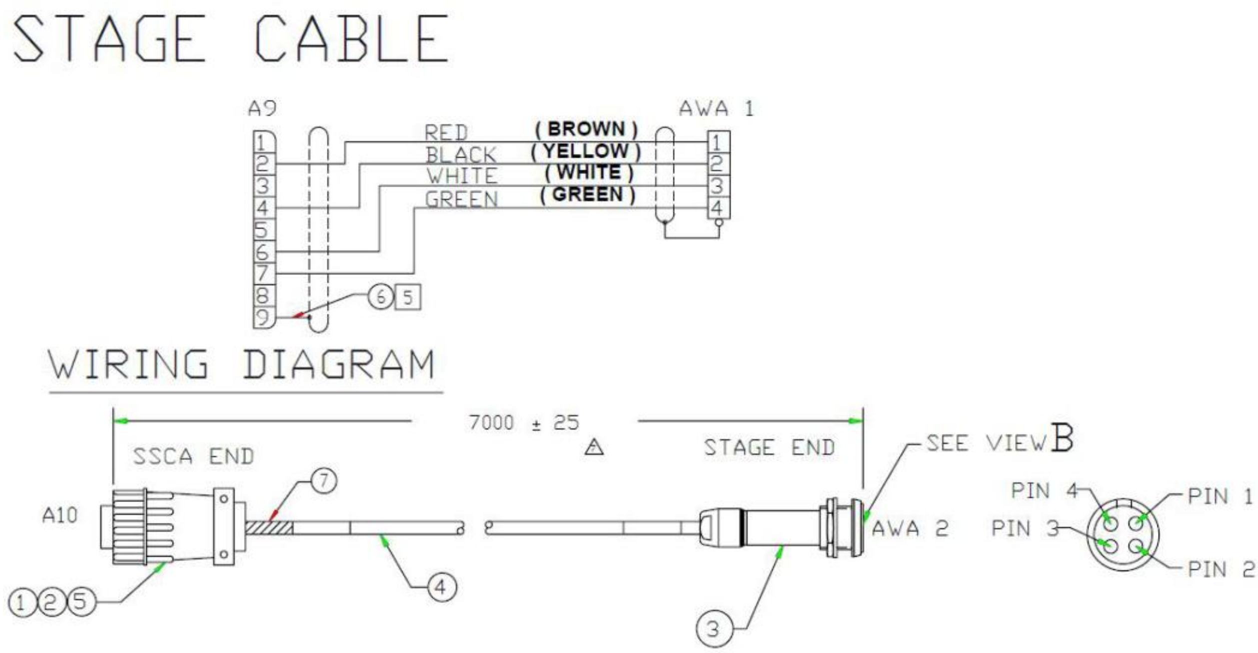


Figure 29 Cable Drawing

32. May refer to below diagram for Right Stepper Motor Cable Routing.

Route : Right Stepper Motor **->** AWA E-Chain -> X axis E-Chain -> Y axis Wide E-Chain -> SSCA

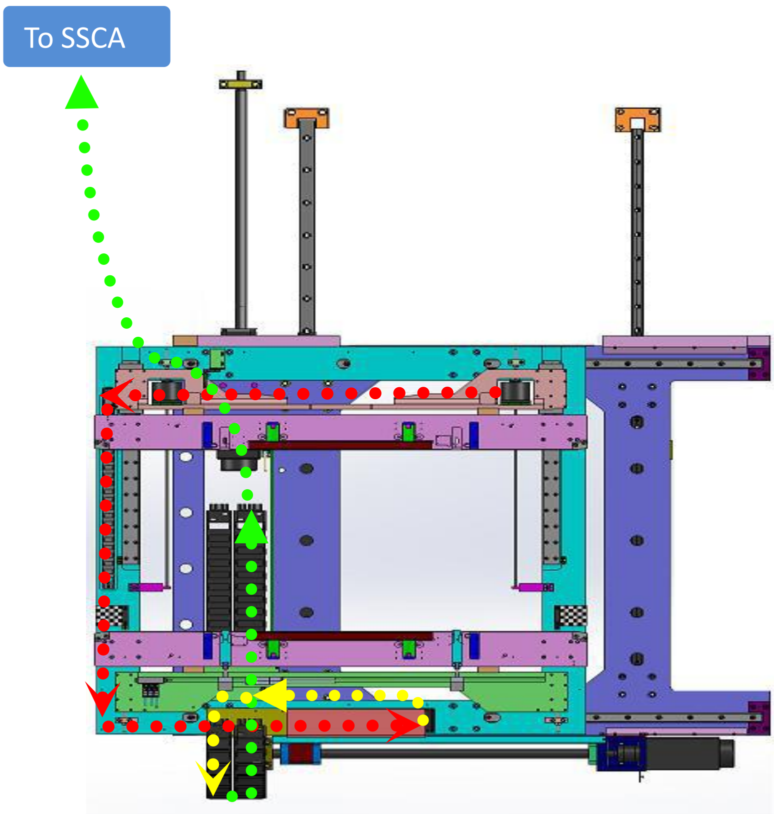


Figure 30 Stepper Motor Cable Routing

33. Assemble Stepper Motor Cable to holding bracket by refer to below Red Dot Orientation .Tighten lock nut by using 17 mm wrench.

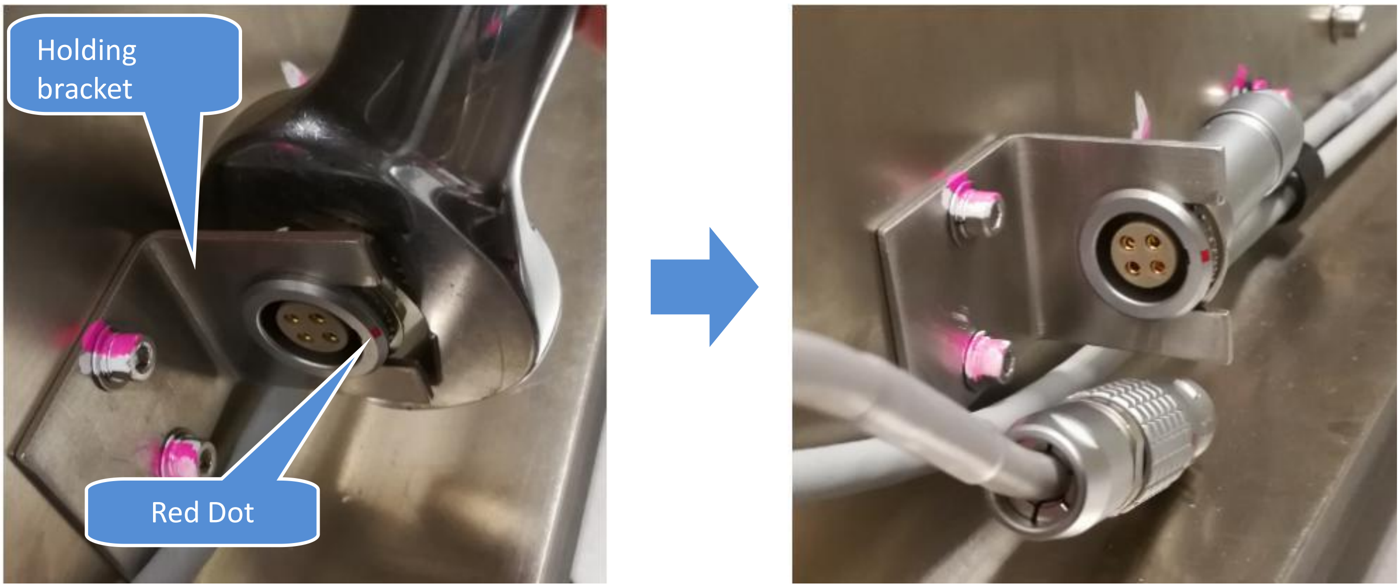


Figure 31 Stepper Motor Cable Assembly

34. Gently route the Stepper Motor Cable as below Figure 32. Secure the cable with Cable Clamp as well.

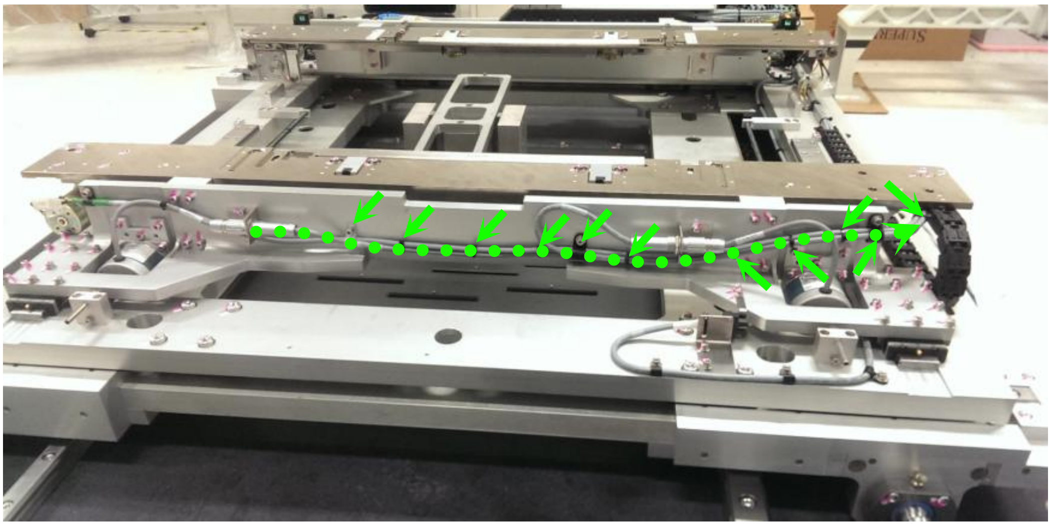


Figure 32 Stepper Motor Cable Routing

35. Connect AWA Stepper motor with 4XASC-A079 respectively by refer to Red Dot Orientation

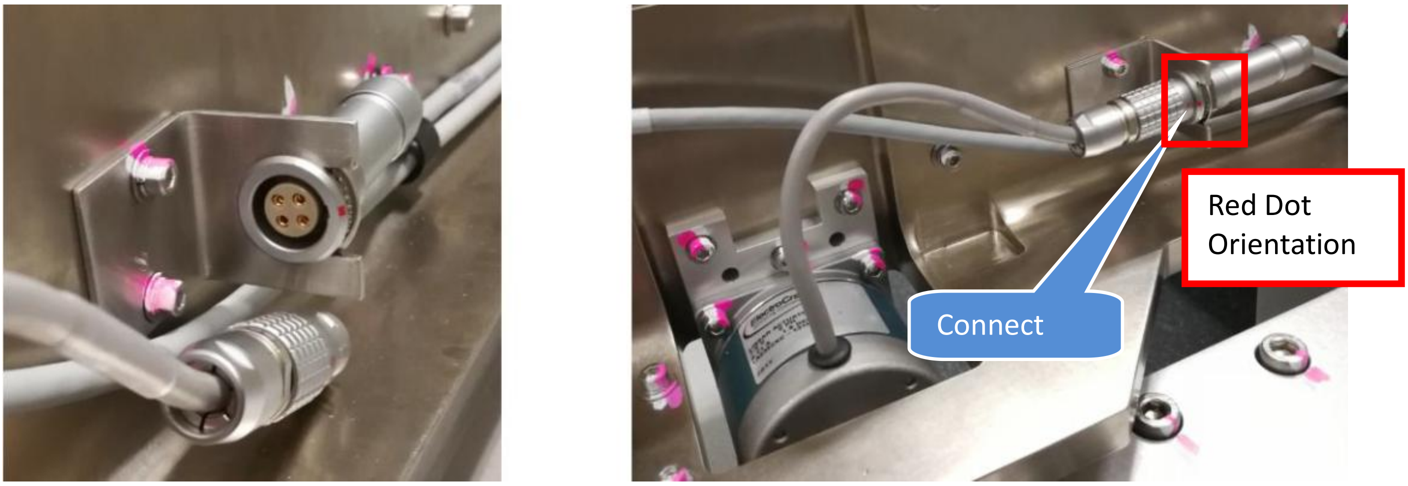


Figure 33 Stepper Motor Cable Connect

36. Route the cable with U shape curve into E-Chain as shown.

    Remarks : Minor tolerance will require to apply to prevent cable snapping

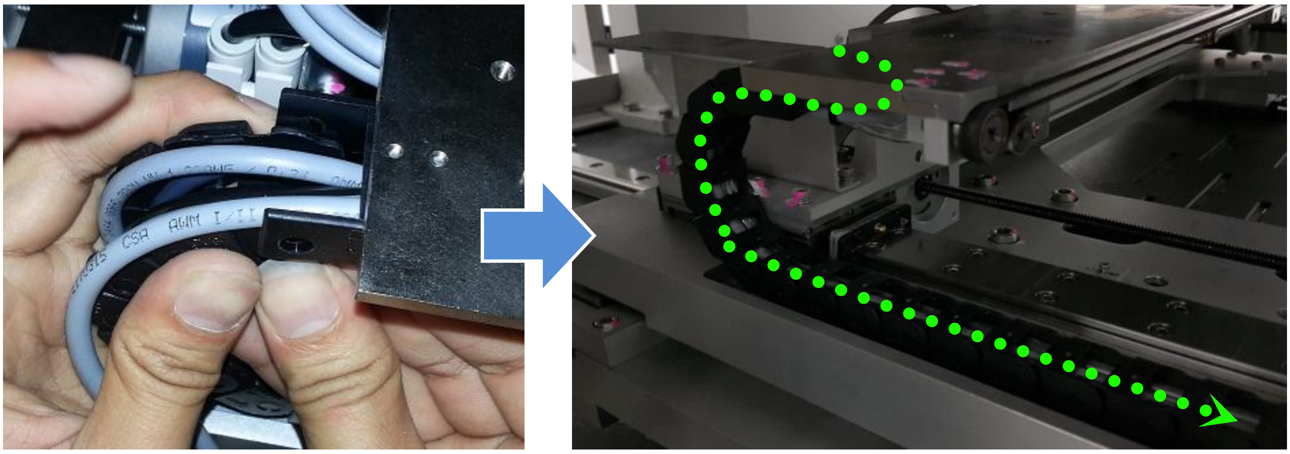


Figure 34 Stepper Motor Cable Routing (AWA E-Chain)

37. Assemble E-chain Top cover at AWA E-Chain. Next,reconnect the E-Chain Top Joint. Secure the AWA Cable with cable tie as below Figure 35

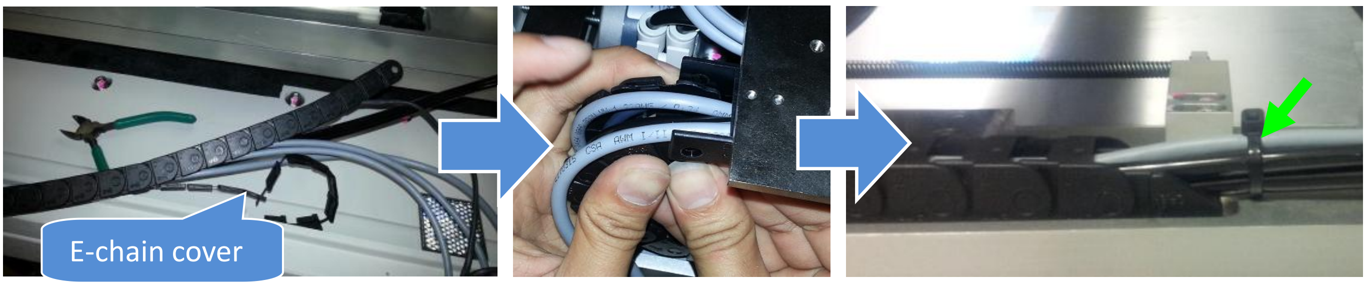


Figure 35 Cable Routing (AWA E-Chain)

38. Assemble 4XAXY-117 to Adj Rail. Tighten 4 x Screws accordingly

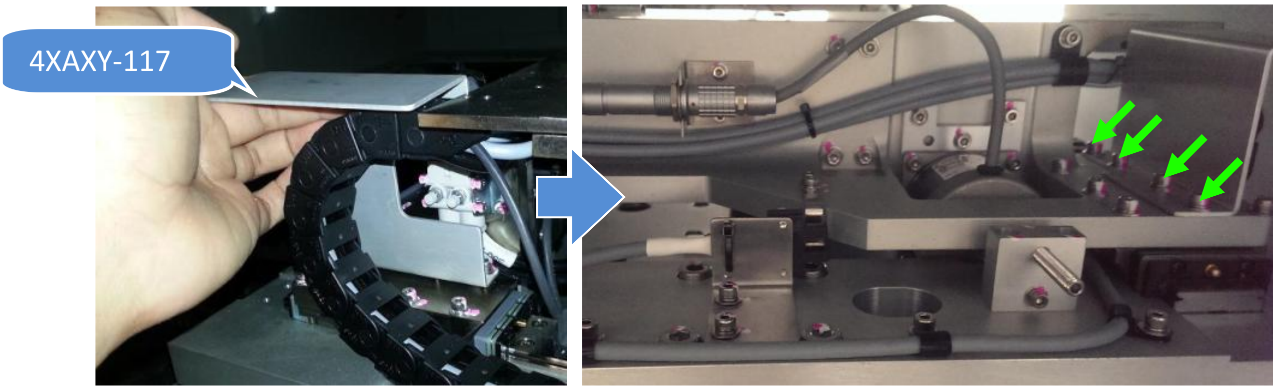


Figure 36 E-Chain Bracket Assembly

39. Gently route the cable as below Figure 37.Secure cable with Cable Clamp & Cable Tie.

Remarks : Caution on Set Screws which protrude at Fix Rail

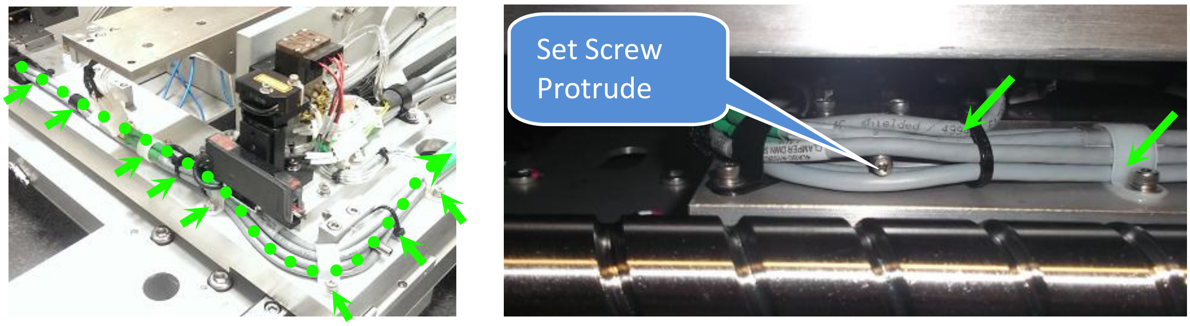


Figure 37 Stepper Motor Cable Routing (Fix Rail)

40. Assemble SENSOR, REFLECTOR, E39-R (P/N, 2284-0052) as below Figure 38

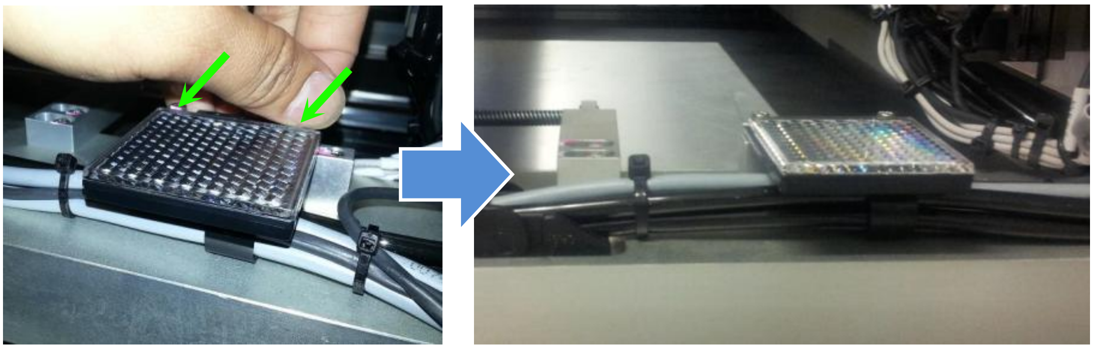


Figure 38 PCL Sensor Reflector Assembly

41. Gently route Stepper Motor Cable into X axis E-Chain by refer to below criteria :
    - ➢ Individual separation layout
    - ➢ Cable do not get caught each other in the shelf parts
    - ➢ Masking Tape MUST remove

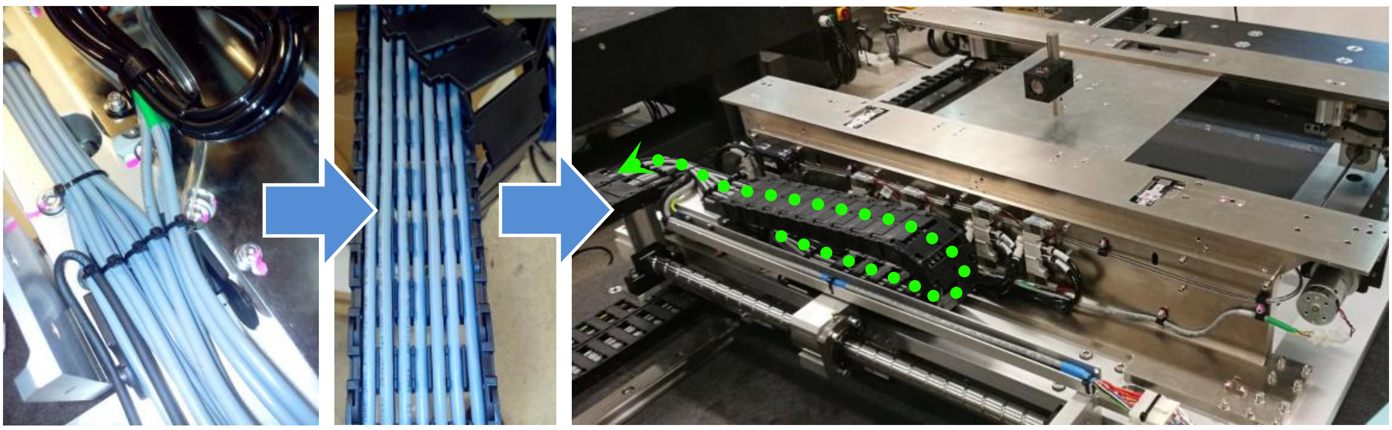


Figure 39 Stepper Motor Cable Routing (X Axis E-Chain)-Reference Picture Only

42. Gently push BRACKET, ENCODER CABLE (P/N, 4XAXY-076) towards rear side .Tighten 2x screws, & reassemble E-Chain cover. Secure Cable with cable tie as below Figure 40

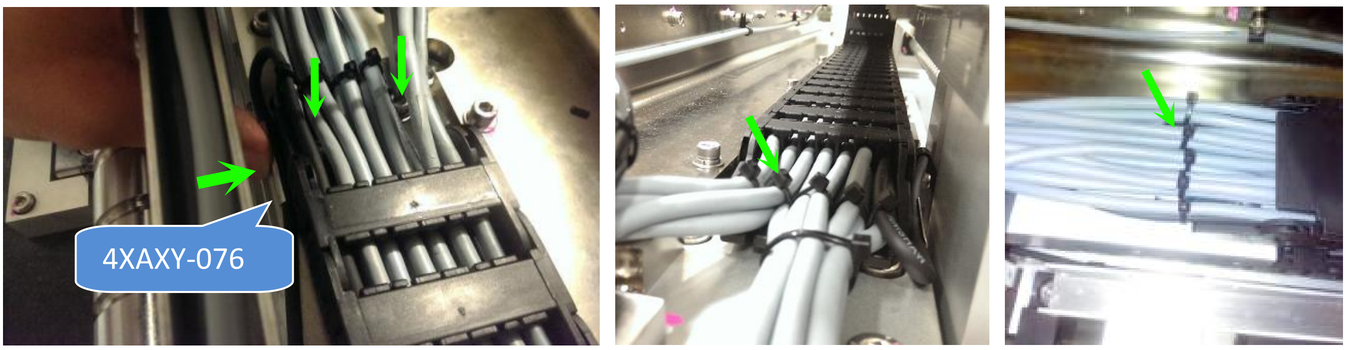


Figure 40 Stepper Motor Cable Routing (X Axis E-Chain)

43. Gently move X axis stage and visual check below item :
    - ➢ X axis Encoder cable do not contain friction with encoder bracket
    - ➢ Valve Assembly (Cable & Tubing) do not contain friction with X Axis E-Chain

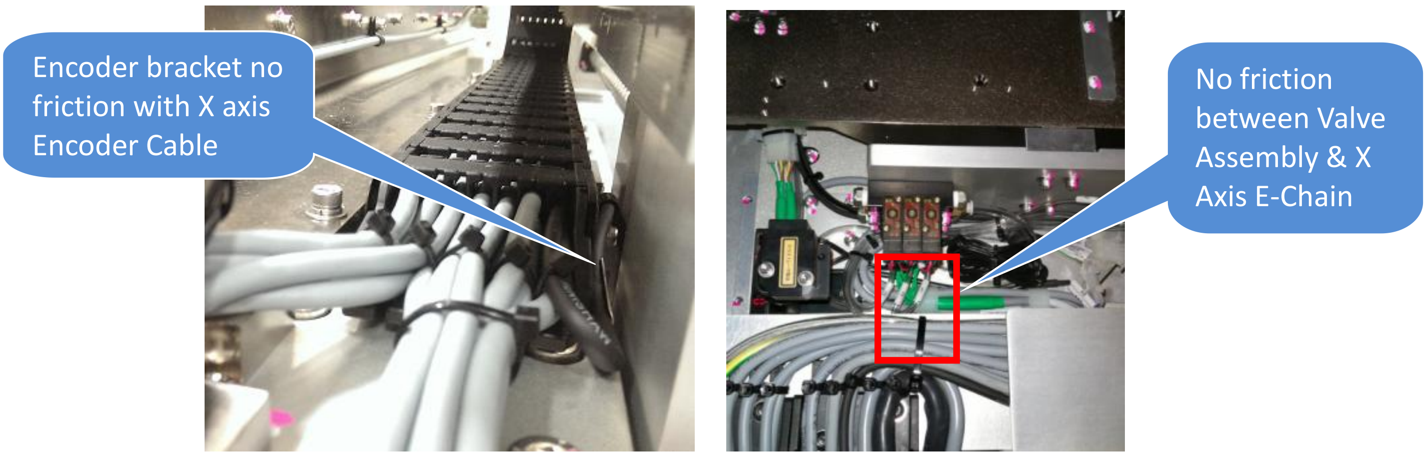
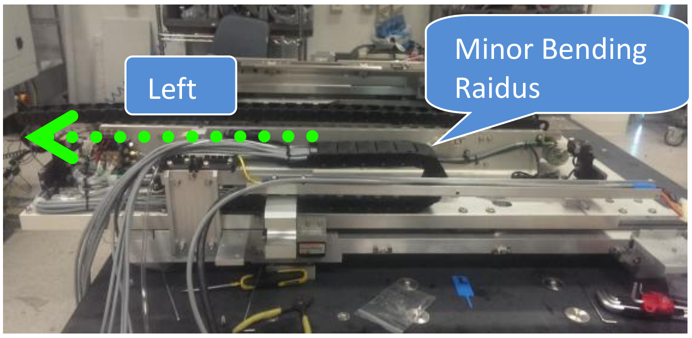


Figure 41

44. Manually move X axis stage towards left & right rapidly and visual check X axis E-chain where do not contain excessive bending radius

Remarks : Re-route the cable if excessive bending radius addressed



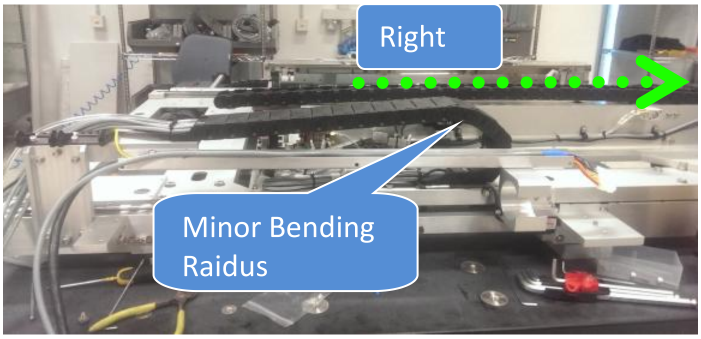


Figure 42 Minor Bending Radius (X axis E-Chain)

45. Assemble 4XAXY-075 as below Figure 43. Secure cables with cable tie

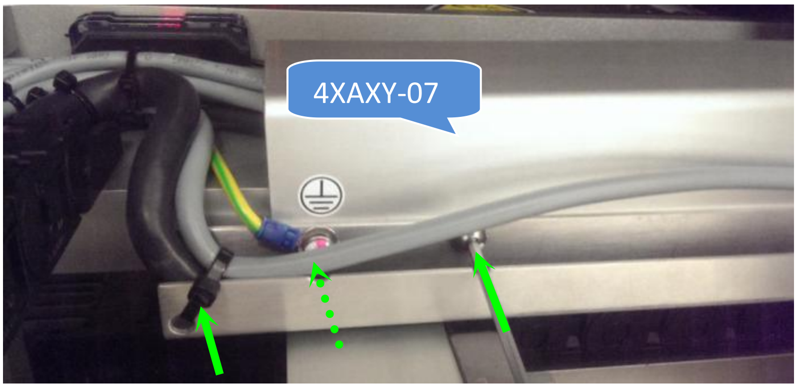


Figure 43 X Axis Guard Plate assembly

46. Route Stepper Motor cable at Y axis Wide E-Chain. Reassemble E-Chain cover.Connect the joint and secure cables with cable tie.

Remarks : Cable Routing Criteria -

- Individual separation layout
- Cable do not get caught each other in the shelf parts
- Masking Tape MUST remove

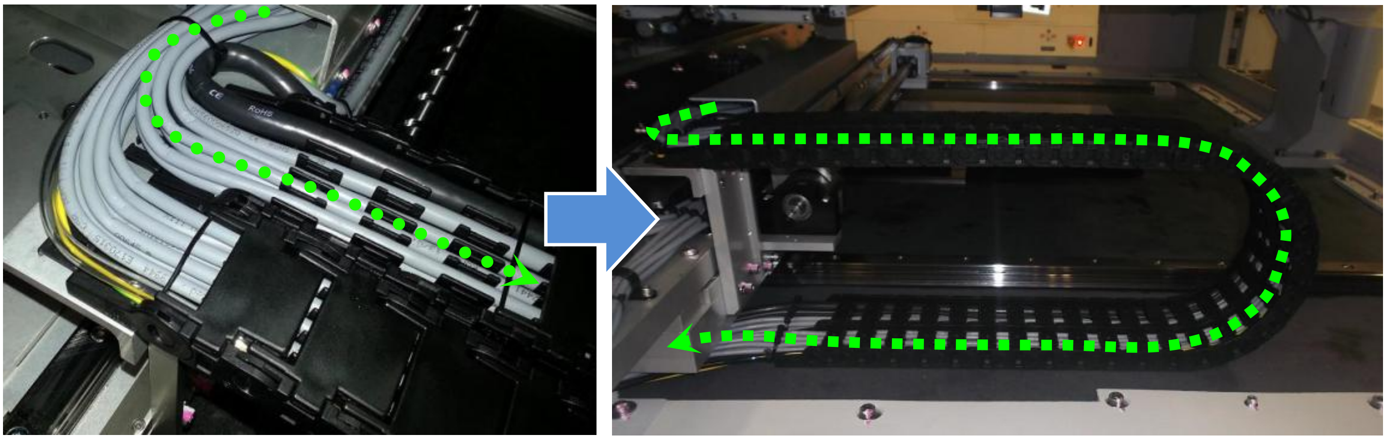


Figure 44 Stepper Motor Cable (Y axis Wide E-Chain)

47. Manually move Y axis stage towards Front & Rear rapidly and visual check Y axis E-chain where do not contain excessive bending radius.

Remarks : Re-route the cable if excessive bending radius addressed

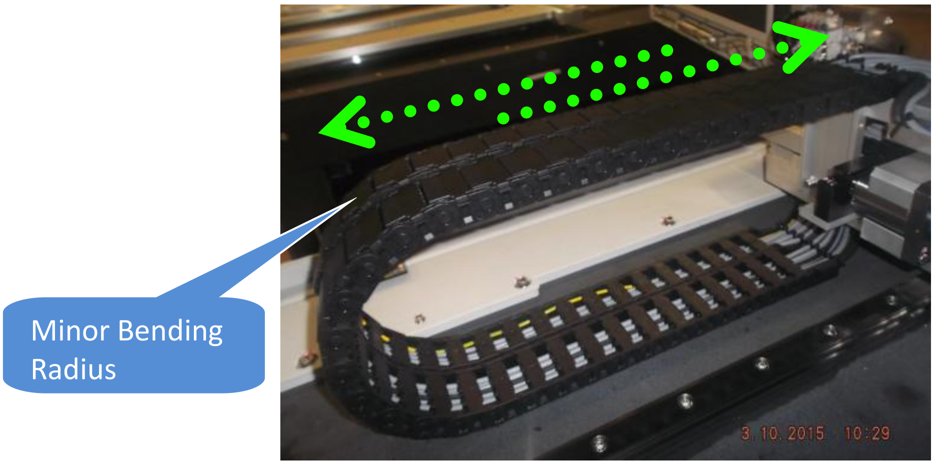


Figure 45 Stepper Motor Cable Routing (Y Axis Wide E-Chain)

48. Route 4XASC-A079 as shown and secure with necessary cable ties.

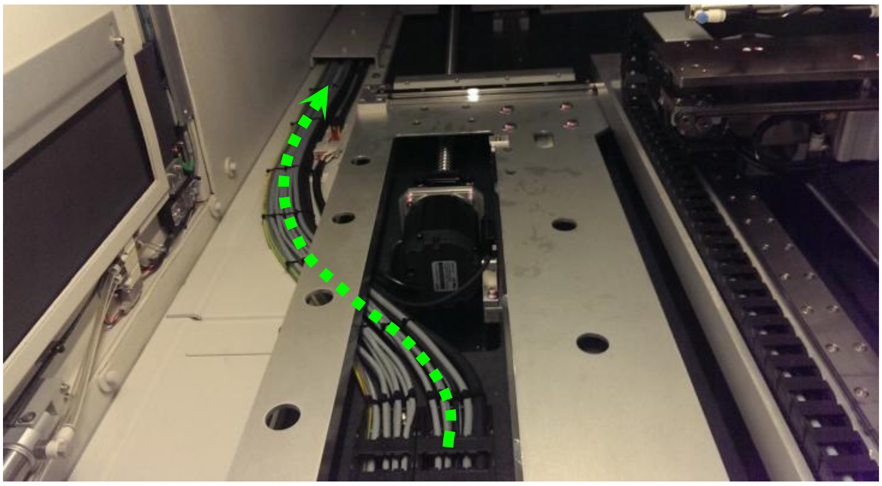


Figure 46 Stepper Motor Cable Routing (Y Axis Wide E-Chain)

49. Gently route 4XASC-A079 through the “Opening” as below Figure 47

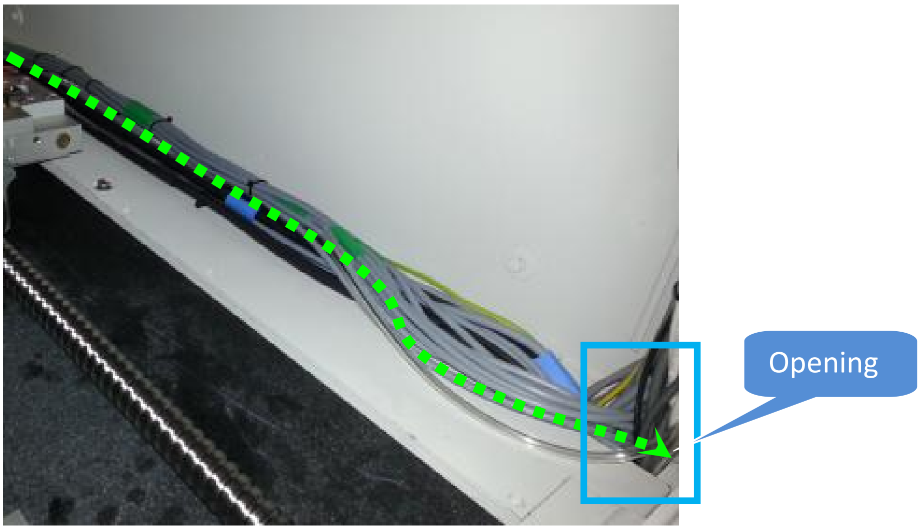


Figure 47

50. Access from rear side,assemble Vertical Lead Shielding Assembly by tighten 2 x screws.Next, assemble Cable Exit Shielding Assembly by tighten 3 x Screws.

Remarks :

- Caution on cable handling upon shielding assembly
- Gaping between both shielding assembly is not allow

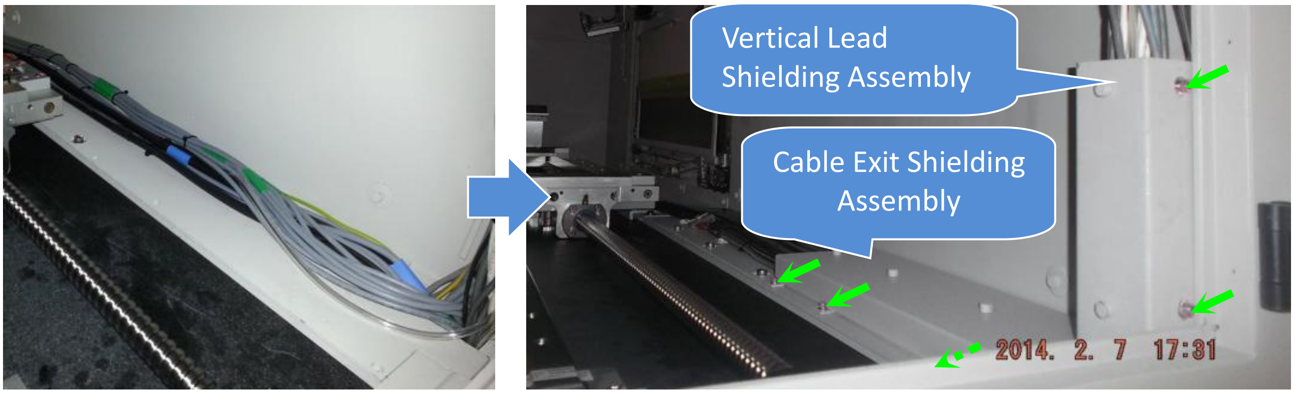


Figure 48 Shielding assembly

51. Route 4XASC-A079 as shown and apply necessary cable ties

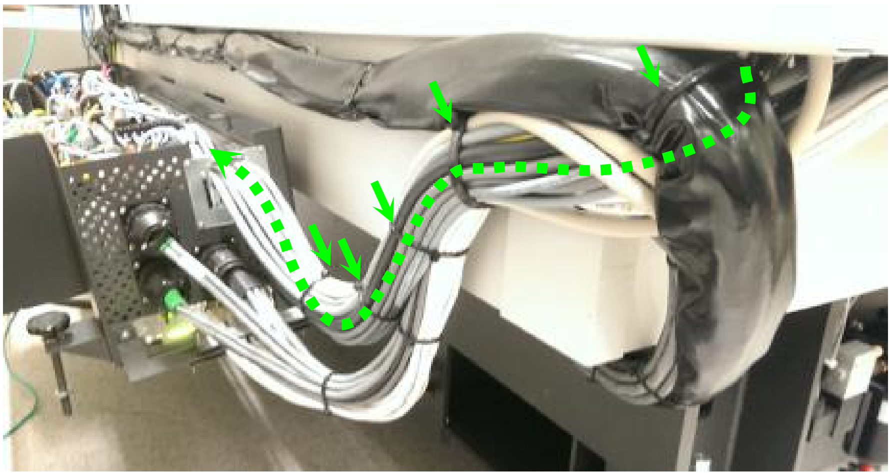


Figure 49 Stepper Motor Cable Routing (SSCA)

52. Route 4XASC-A079 as shown

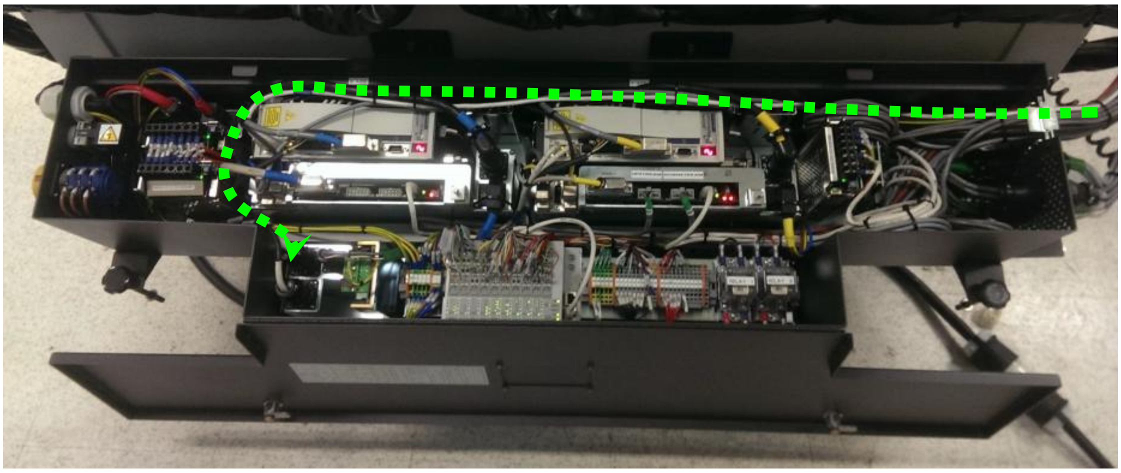


Figure 50 Stepper Motor Cable Routing (SSCA)

53. Connect end bulb head connector to AWA stepper driver

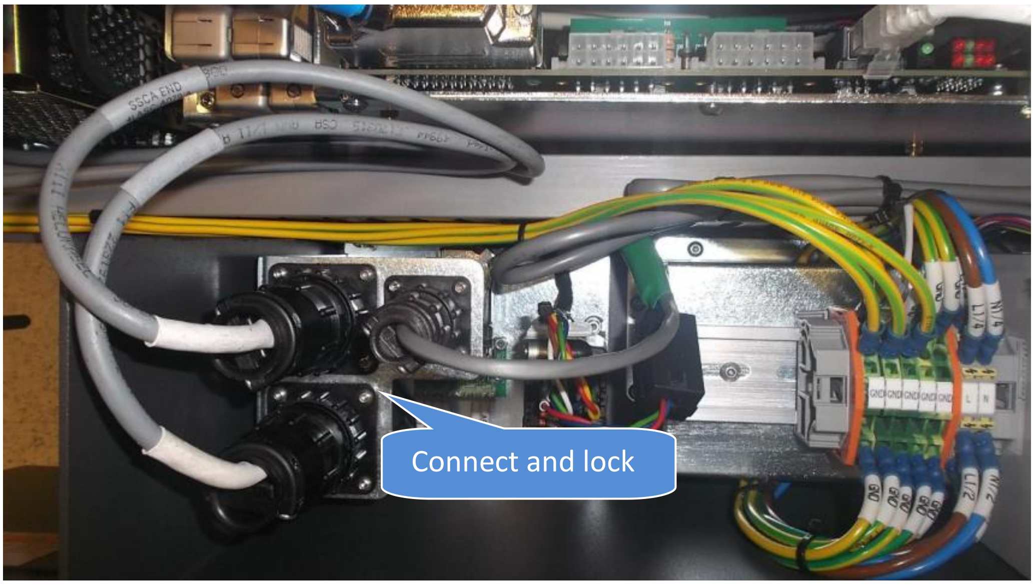


Figure 51 Stepper Motor Bulk Head Connector

54. Reassemble 4XAMC-A045 as shown. Tighten 2 x Screws accordingly

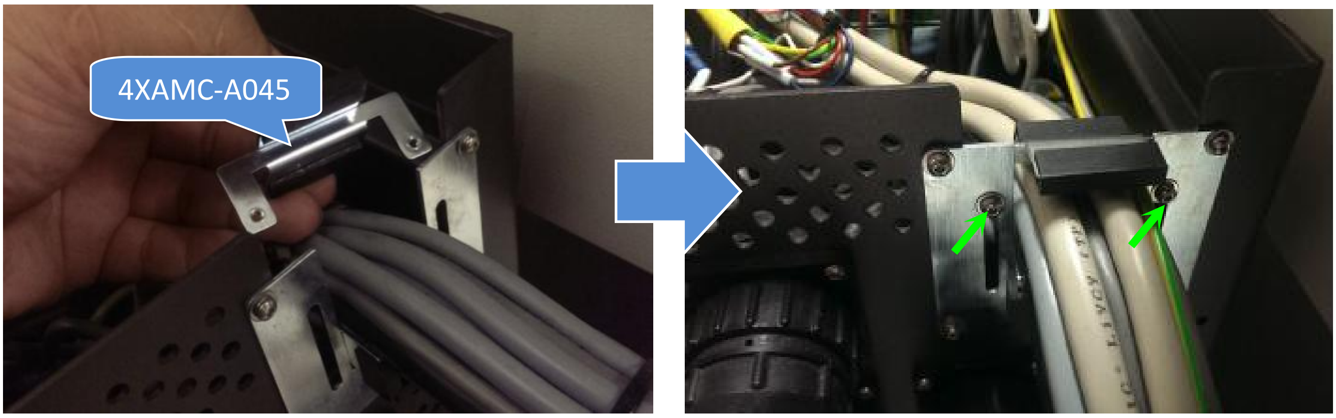


Figure 52 Holding Plate Assembly

55. Gently assemble Right Outer Barrier.Tighten 4 x Tamper Proof screws as below Figure 53

Figure 53 Right Outer Barrier Assembly

56. Connect 4 x connectors at Right Outer Barrier

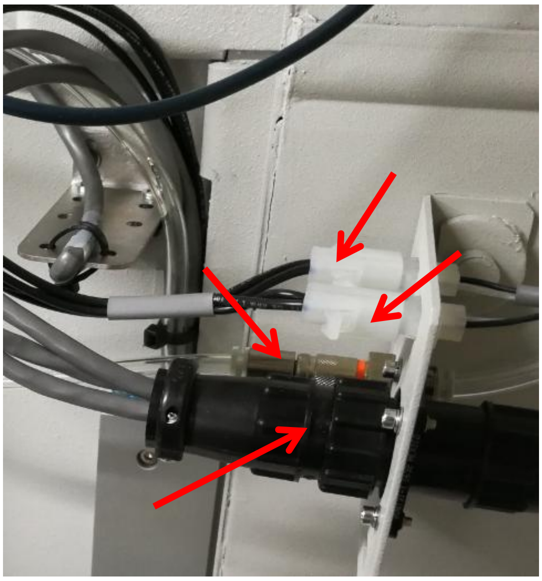


Figure 54 Outer Barrier Connector Assembly

57. Turn On SSCA

58. Perform X ray Survey

AWA Rail Width Validation

59. Close all door access.Start-up machine as usual.

60. Log in V810 GUI and initialize DIO.

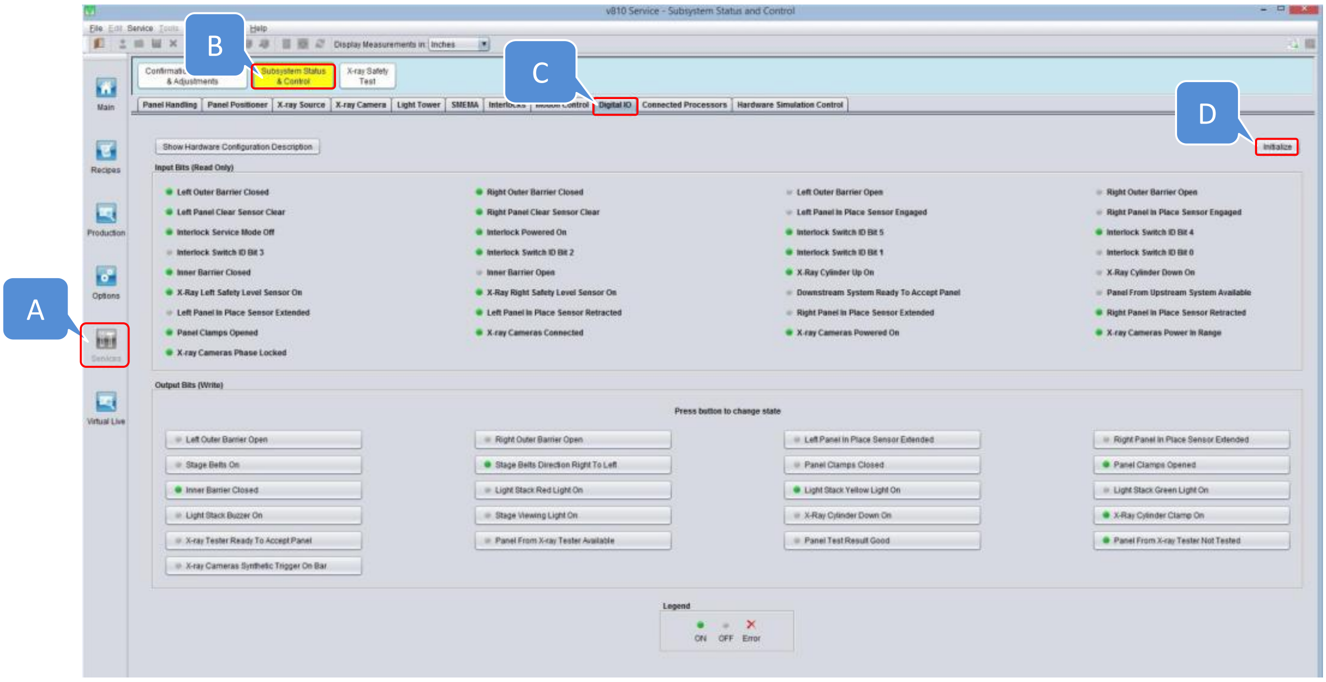


Figure 55 Digital IO Tab

61. Go to Panel Handling ,change display measurement set as **"MM"**.Click initialize and manually set rail width into 150 **mm**

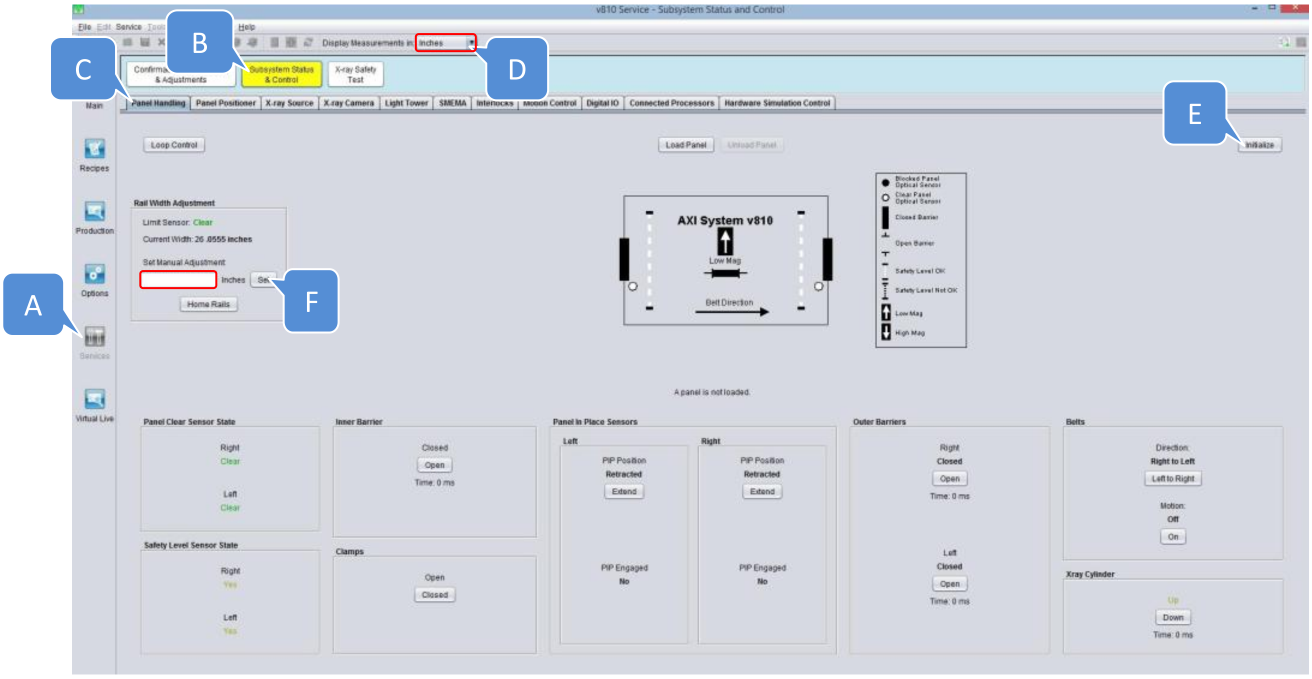


Figure 56 Set Width 150MM

62. Turn off X-ray source at Operator Console Panel
63. Open front access door
64. Gently pull the XY Stage towards front door direction
65. Place Caliper as below position. (May refer to below Figure 57)

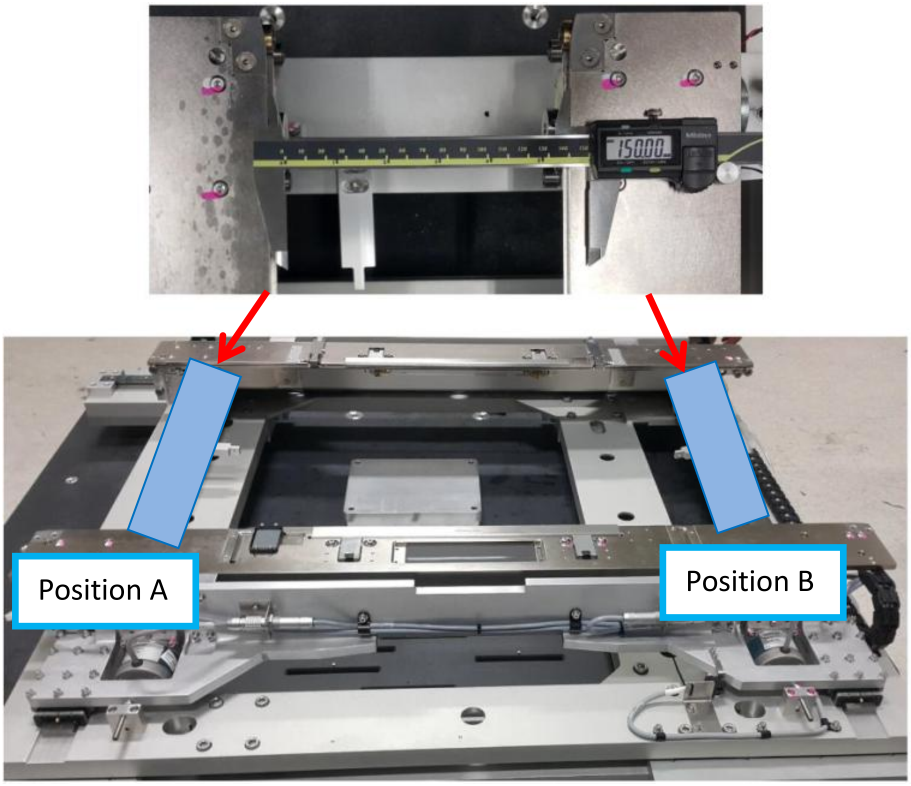
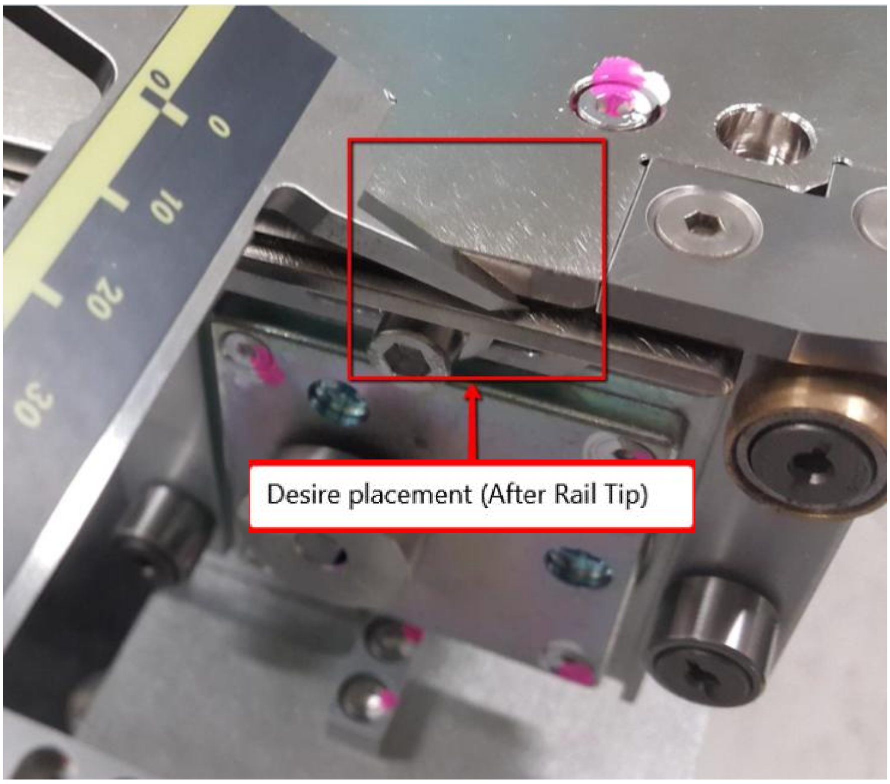
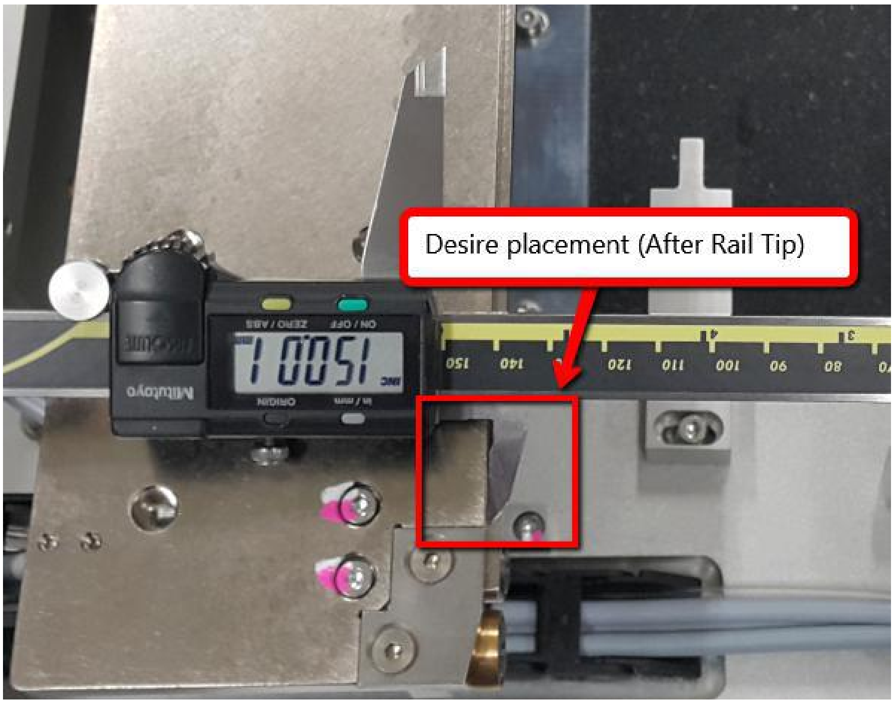


Figure 57 Caliper Placement

Position	Position A & Position B
Fix rail Caliper placement	
Adj rail Caliper placement	

66. Record 5 repeated measurements at the right end (Position A), and then 5 repeated measurements at the left end (Position B):
    - MRave = average value of 5 repeated measurements at right end of conveyor
    - MLave = average value of 5 repeated measurements at left end of conveyor
67. Calculate the parallelism, Mave and width error using the formulas below. Compare these values to the parallelism and width error limits.
    - Parallelism = MRave - MLave
    - Mave = (MRave + MLave)/2
    - Width Error = Set Width - Mave
    - Limits on Parallelism = +/- 0.125mm
    - Limits on Width Error = +/- 0.125mm
68. By refer to below table and step, check Parallelism Limit & Width Accuracy Limit

Sample Measurement (mm)			
Measurement Right End (Position A)		Measurement Left End (Position B)	
MR1	150.00	ML1	150.10
MR2	150.01	ML2	150.08
MR3	150.05	ML3	150.08
MR4	150.00	ML4	150.06
MR5	150.01	ML5	150.09

Calculations:

Measurement Right End average (MRave) = (MR1+MR2+MR3+MR4+MR5)/5 = 150.01

Measurement Left End average (MLave)= (ML1+ML2+ML3+ML4+ML5)/5 = 150.08

Measurement average = (MRave + MLave)/2 = 150.04

Compare to Limits for Parallelism:

MRave – MLave = **0.07 is less than Spec "+/- 0.125mm ",** so no adjustment for parallelism is required

Compare to Limits for Width Accuracy:

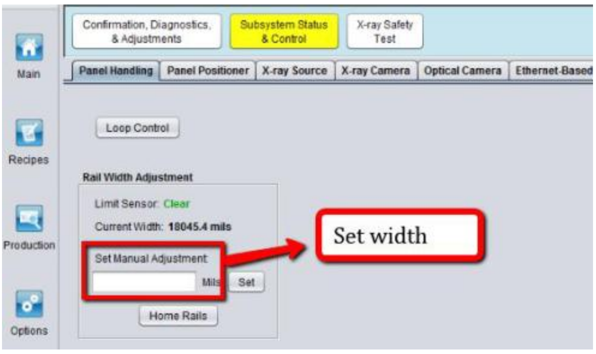


Figure 3 Set Width

Set Width – Mave = **0.04 is less than Spec "+/- 0.125mm ",** so no adjustment required for width accuracy

69. If not within spec, may refer to AXI V810 STD AWA Rail Width Adjustment Procedure for optimization.